Experiment 10

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1:
Set Up Your HTML File – Create an HTML file (index.html) with two canvas elements for the pie chart and bar chart. Include inline CSS to center the layout and add spacing. Link Chart.js via CDN and reference the script.js file.

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.
    <title>Inventory Management Visualization</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            margin: 50px;
        }
        canvas {
            margin: 20px auto;
        }
    </style>
</head>
<body>
    <h1>Inventory Management System</h1>
    <canvas id="pieChart" width="400" height="400"></canvas>
    <canvas id="barChart" width="400" height="400"></canvas>

    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <script src="script.js"></script>
</body>
</html>
```

index.html – HTML structure with two canvas elements and Chart.js CDN link

Step 2:

Create the JavaScript File for Charts – Create a script.js file to handle the data visualization logic. Define inventory data with labels and stock quantities for five categories (Electronics, Clothing, Home Appliances, Books, Toys). Render a Pie Chart titled 'Inventory Distribution' and a Bar Chart titled 'Items in Stock by Category' using Chart.js.

```javascript
// Data for inventory
const inventoryData = {
    labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
    datasets: [{
        label: 'Items in Stock',
        data: [200, 150, 100, 80, 50],
        backgroundColor: [
            '#FF6384', '#36A2EB', '#FFCE56', '#4BC0C0', '#9966FF'
        ]
    }]
};

// PIE Chart
const ctxPie = document.getElementById('pieChart').getContext('2d');
new Chart(ctxPie, {
    type: 'pie',
    data: inventoryData,
    options: {
        responsive: true,
        plugins: {
            title: {
                display: true,
                text: 'Inventory Distribution'
            }
        }
    }
});

// BAR Chart
const ctxBar = document.getElementById('barChart').getContext('2d');
new Chart(ctxBar, {
    type: 'bar',
    data: inventoryData,
    options: {
        responsive: true,
        plugins: {
            title: {
                display: true,
                text: 'Items in Stock by Category'
            }
        },
        scales: {
            y: {
                beginAtZero: true
            }
        }
    }
});
```

script.js – Inventory data definition, Pie Chart and Bar Chart rendering using Chart.js

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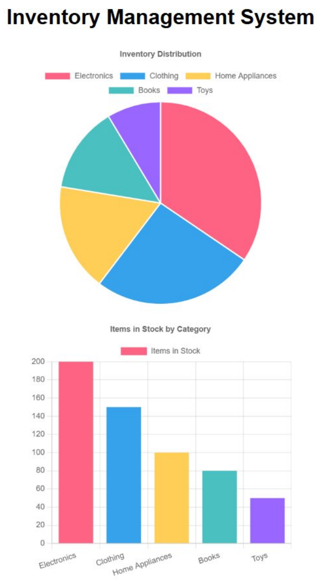
OUTPUT:

The output displays the Inventory Management System page with two charts rendered side by side using Chart.js on an HTML canvas. The Pie Chart titled 'Inventory Distribution' shows the proportional share of each category — Electronics (200), Clothing (150), Home Appliances (100), Books (80), and Toys (50) — each in a distinct color. Below it, the Bar Chart titled 'Items in Stock by Category' visualizes the same data as vertical bars, clearly showing Electronics has the highest stock and Toys the lowest.



Output – Inventory Management System showing Pie Chart and Bar Chart rendered in browser

Chart Details:

- **Pie Chart** – Shows inventory distribution across all 5 categories proportionally.
- **Bar Chart** – Shows items in stock per category with Y-axis starting at zero.
- **Data:** Electronics: 200 | Clothing: 150 | Home Appliances: 100 | Books: 80 | Toys: 50
- **Colors:** #FF6384 (pink), #36A2EB (blue), #FFCE56 (yellow), #4BC0C0 (teal), #9966FF (purple)

Result: Thus, data visualizations including a pie chart and a bar graph for an inventory management system were successfully created using HTML, CSS, and JavaScript with Chart.js.